

MBS INTL, MI, USA

WMO: 726379

Lat: 43.533N

Lon: 84.080W

Elev: 660

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14845

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	4.1	-7.9	3.7	1.4	-3.3	4.7	5.3	27.3	23.3	25.3	25.6	8.6	260	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.4	73.1	86.4	71.3	83.7	70.1	76.0	85.2	74.2	82.8	72.5	80.4	12.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	125.9	81.1	71.6	119.4	79.1	69.9	112.6	77.6	39.8	85.8	38.1	82.7	36.5	80.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.5	19.3	DB	-5.5	94.4	6.4	3.0	-10.1	96.6	-13.9	98.3	-17.5	100.0	-22.2	102.2	
			WB	-6.1	79.4	6.3	2.2	-10.6	81.0	-14.3	82.3	-17.9	83.6	-22.5	85.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	22.8	24.6	33.9	45.9	58.1	68.1	71.6	69.6	62.9	51.2	39.0	28.8	(6)
	DBStd	19.33	10.75	10.41	10.69	8.83	8.67	7.07	5.63	5.38	7.35	8.61	8.86	9.35	(7)
	HDD50	3370	845	713	514	179	20	0	0	0	3	92	346	658	(8)
	HDD65	6774	1309	1133	965	574	254	47	9	19	125	437	780	1122	(9)
	CDD50	2694	0	0	14	57	270	543	670	606	389	128	16	1	(10)
	CDD65	625	0	0	1	2	39	139	214	160	62	8	0	0	(11)
	CDH74	5389	0	0	9	41	388	1268	1884	1189	532	78	0	0	(12)
	CDH80	1605	0	0	2	6	107	426	606	306	137	15	0	0	(13)
	WSAvg	9.4	10.7	10.6	10.4	11.0	9.7	8.5	7.6	7.1	7.7	9.3	10.2	10.3	(14)
	PrecAvg	31.1	1.7	1.5	2.4	2.9	2.8	2.4	3.3	3.9	2.4	2.6	2.2	2.2	(15)
(16) Precipitation	PrecMax	39.8	4.1	6.1	6.7	6.9	6.2	5.2	5.3	9.0	16.1	5.6	5.8	5.3	(16)
	PrecMin	21.3	0.6	0.2	0.3	1.1	0.8	0.6	0.4	0.6	0.0	0.5	0.5	0.6	(17)
	PrecStd	4.5	1.0	1.2	1.3	1.3	1.3	1.1	1.3	1.9	2.8	1.2	1.3	1.2	(18)
	PrecStd	4.5	1.0	1.2	1.3	1.3	1.3	1.1	1.3	1.9	2.8	1.2	1.3	1.2	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	56.1	72.7	80.0	87.7	92.6	93.2	90.6	88.9	81.7	67.3	57.9	(19)
		MCWB	50.1	50.7	60.3	62.9	68.6	74.1	75.9	74.5	71.2	67.2	57.3	53.7	(20)
	2%	DB	45.7	47.6	63.1	72.5	83.1	88.5	89.4	86.3	83.8	75.0	61.2	50.0	(21)
		MCWB	42.8	43.6	54.8	58.2	67.0	71.2	74.6	72.5	69.2	63.5	54.2	46.6	(22)
	5%	DB	40.1	42.1	55.4	67.0	78.5	85.2	86.1	83.2	80.4	70.0	56.8	45.1	(23)
		MCWB	37.6	38.3	48.8	54.5	64.0	69.9	72.1	71.1	67.7	61.2	51.2	41.9	(24)
	10%	DB	36.5	37.9	49.1	61.9	73.5	81.3	83.2	80.7	76.4	65.5	52.4	40.6	(25)
		MCWB	34.6	34.7	43.3	51.9	61.9	68.2	70.3	69.3	65.6	58.3	47.7	37.6	(26)
	0.4%	WB	50.7	51.4	61.8	66.1	72.5	76.4	79.4	77.7	75.1	70.3	59.5	54.7	(27)
		MCDB	52.1	54.1	72.7	76.3	83.1	88.6	89.5	87.1	84.5	78.1	64.5	58.0	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.0	44.0	55.4	60.4	69.1	74.0	76.8	75.0	72.1	66.2	55.6	47.5	(29)
		MCDB	45.3	47.7	61.5	68.5	78.1	85.0	85.8	82.7	79.3	71.8	60.3	49.6	(30)
	5%	WB	37.9	38.6	49.5	56.7	66.5	72.1	74.7	73.2	69.8	62.3	51.8	42.3	(31)
		MCDB	39.6	41.6	54.8	65.0	75.4	81.7	83.1	80.4	76.3	68.9	56.2	45.0	(32)
	10%	WB	34.6	35.1	43.5	52.7	63.6	70.1	72.7	71.3	67.6	58.9	47.6	37.7	(33)
		MCDB	36.4	37.7	48.9	61.2	72.0	78.1	80.1	78.0	74.3	64.7	51.9	40.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	13.8	15.9	19.0	19.7	19.6	19.8	19.3	20.8	17.8	13.7	11.1	(35)
		MCDBR	15.1	16.5	24.4	27.7	24.7	23.2	22.5	21.7	25.0	24.6	19.3	15.5	(36)
		MCWBR	13.7	13.6	17.0	16.5	12.8	10.5	10.8	10.5	12.3	14.4	14.5	13.4	(37)
	5% WB	MCDBR	14.0	15.9	23.1	24.8	21.5	19.9	19.2	18.3	20.1	22.2	18.1	15.2	(38)
		MCWBR	13.0	13.8	17.2	17.2	12.9	10.9	10.5	10.2	11.5	14.3	14.9	13.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.291	0.317	0.331	0.365	0.390	0.419	0.407	0.400	0.374	0.341	0.311	0.284	(40)
	taud		2.379	2.304	2.386	2.317	2.302	2.252	2.292	2.306	2.377	2.462	2.492	2.446	(41)
	Ebn at Noon		266	277	288	285	280	271	273	271	270	266	259	259	(42)
	Edh at Noon		25	32	33	39	40	43	41	39	33	27	22	21	(43)
(44) All-Sky Solar Radiation	RadAvg		443	725	1137	1450	1705	1916	1931	1645	1312	822	497	353	(44)
	RadStd		43	90	121	124	153	108	96	73	125	83	64	32	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page